A STUDY ON IMPACT OF PREDICTIVE ANALYSIS FOR COMPANIES ENGAGING PERFORMANCE AFTER THE PANDEMIC IN BBK EXPORTS PRIVATE LIMITED, KILAMBI.

A Project report

submitted by

T. VIJAYALAKSHMI RASATHI

(Register No. 513224631059)

under the guidance of

M. SHIYAMALA, MBA.,

Faculty of Master of Business Administration in

Partial fulfillment of the requirements

For the award of the degree of

MASTER OF BUSINESS ADMINISTRATION

IN

BUSINESS ANALYTICS



ANNA UNIVERSITY

CHENNAI - 600 025

JULY - 2025

BONAFIDE CERTIFICATE

Certified that this project report titled on **"A STUDY ON IMPACT OF PREDICTIVE ANALYSIS FOR COMPANIES ENGAGING PERFORMANCE AFTER THE PANDEMIC IN BBK EXPORTS PRIVATE LIMITED"**, is submitted by **VIJAYALAKSHMI RASATHI T (Reg.No. 513224631059)** who carried out the research under my supervision. Certified further, that to the best of my knowledge, the work reported here is does not from part of any other report or dissertation on the basis of which a degree or award was conferred on an earlier on this or any other candidate.

Dr. R. MUGUNDHAN, M.Com.,
M.Phil., LLB., Ph.D.
DIRECTOR OF THE DEPARTMENT
P.G. Dept. of Management Studies
Thirumalai Engineering College
Kilambi - 631 551.

M. SHIYAMALA, MBA.,
ASSISTANT PROFESSOR
P.G. Dept. of Management Studies
Thirumalai Engineering College,
Kilambi – 631 551.

VIVA VOCE EXAMINATION HELD ON

INTERNAL EXAMINER **EXTERNALEXAMINER**

DECLARATION

I hereby declare that the project work entitled "**A STUDY ON IMPACT OF PREDICTIVE ANALYSIS FOR COMPANIES ENGAGING PERFORMANCE AFTER THE PANDEMIC IN BBK EXPORTS PRIVATE LIMITED, KILAMBI"** in partial fulfillment of Master of Business Administration course under Anna University is an original work done by me under the guidance of **M. SHIYAMALA, MBA.,** Assistant professor, Thirumalai Engineering College, Kilambi, Kanchipuram.

Place: VIJAYALAKSHMI RASATHI T

Date: (513224631059)

ACKNOWLEDGEMENT

First I would like to thank the almighty for being with me and forgiving me the courage to pursue.

It gives me great pleasure to express my boundless sense of gratitude to each and every person who directly or indirectly helped me with hand in hand completing this humble piece of work.

I take this opportunity to thank all those who were of great support in finishing the report successfully.

I wish to express my sincere gratitude to the principal of our college **Mr. Dr. S. PRABHAKARAN, and Mr. Dr. R. MUGUNTHAN,** Director of the department, and **M. SHIYAMALA,** faculty guide for their guidance and encouragement in carrying out this project report. I thank the department of business administration in Thirumalai Engineering College, for providing me with an opportunity to gain practical exposure to the outside world.

I also wish to thank **Mr. DEVANANDHAN. G (HR MANAGER)** of **BBK Exports Private Limited** for providing me an opportunity to do my internship and guiding me. I also thank **Mr. RANJITH KUMAR. J (HR**) teaching me about the work in the companies, and also thank officials of BBK Exports private limited who rendered their help during the period of project work.

TABLE OF CONTENT

SI.NO	TOPIC	PAGE NO
CHAPTER - I	**INTRODUCTION**	
	INTRODUCTION	**01**
	OBJECTIVES OF THE STUDY	**04**
	NEEDS OF THE STUDY	**05**
	SCOPE OF THE STUDY	**06**
	LIMITATION OF THE STUDY	**07**
	COMPANY PROFILE	**10**
	PRODUCT PROFILE	**16**
CHAPTER – II	**REVIEW OF LITERATURE**	
	LITERATURE SURVEY	**18**
CHAPTER – III	**RESEARCH METHODOLOGY**	
	RESEARCH DESIGN	**23**

	DATA COLLECTION	**24**
	TOOLS AND TECHNIQUES	**25**
CHAPTER – IV	**DATA ANALYSIS AND INFERENCE**	
	DATA ANALYSIS AND INFERENCE	**26**
CHAPTER – V	**SUMMARY OF FINDINGS, SUGGESTIONS AND CONCLUSION**	
	FINDINGS	**42**
	SUGGESTIONS	**43**
	CONCLUSION	**44**
	APPENDIX	
	QUESTIONNARIE	**45**
	BIBLIOGRAPHY	**50**

LIST OF CHARTS AND TABLES

CHARTS AND TABLES NO	TITLE	PAGE NO
01	AGE	**26**
02	GENDER	**27**
03	EDUCATION QUALIFICATION	**28**
04	WORK EXPERIENCE	**29**
05	AWARE OF PREDICTIVE ANALYSIS	**30**
06	DEPARTMENT USING PREDICTIVE ANALYSIS	**31**
07	IMPROVED COMPANY'S DECISION MAKING	**32**
08	LEARN ABOUT PREDICTIVE ANALYSIS	**33**
09	UNDERSTAND PREDICTIVE ANALYSIS	**34**
10	MANAGER EXPLAIN PREDICTIVE ANALYSIS	**35**
11	HELPS TO AVOID BUSINESS RISKS	**36**
12	FIRST HEAR ABOUT PREDICTIVE ANALYSIS	**37**
13	BUSINESS BENEFITS	**38**
14	UNDERSTAND CUSTOMER NEEDS	**39**
15	IMPROVED CUSTOMER SATISFACTION	**40**
16	USING ANALYTICS TOOLS	**41**

INTRODUCTION
CHAPTER - 1

This is a study on impact of predictive analytics for company's engaging performance after the pandemic. The global business environment has undergone a seismic since the onset of the COVID- 19 pandemic in early 2020. Industries across all sectors, especially those dependent on global trade and logistics, were confronted with unprecedented challenges. Predictive analysis is a type of digital marketing data that is examined with the appropriate digital marketing technologies. This analytics employs machine learning or statistics to forecast future events ranging from sales trends to customer involvement patterns. The purpose is to provide the best judgement of what will happen in the future, rather than simply knowing what has happened.

Predictive analysis refers to the use of data-driven techniques such as statistical algorithms, Artificial Intelligence (AI), and Machine Learning (ML) to forecast future trends, behaviors, and outcomes. By analyzing historical and real- time data, predictive models help businesses anticipate changes, identify opportunities, and mitigate risks. After the pandemic, organization increasingly relied on predictive analysis to recover from financial losses, adapt to changing market condition, and improve overall operational efficiency.

One of the most significant impact of predictive analysis post pandemic its to ability to enhance decision- making. Traditional reactive approaches proved ineffective in uncertain environments. Predictive tools enabled companies to take proactive actions such as forecasting customer demand, optimizing resources allocation, and identifying supply chain risks before they retail sector, predictive analysis allowed businesses to track customer behavior shifts and adjust inventory and marketing strategies accordingly.

Another crucial area where predictive analysis made a major difference was in risk management and financial planning. Companies used forecasting models to assess financial risks, plan budgets effectively, and set realistic business goals based on likely future scenarios. This led to increased stability and better performance tracking even in volatile markets.

The post pandemic era has seen a rapid acceleration in digital transformation, and predictive analysis has become a cornerstone of this shift. It has helped companies not only survive but thrive by improving their agility, adaptability, and strategic performance. As businesses continue to face evolving challenges, predictive analysis will remain a key driver of innovation and long-term success.

MEANING OF COMPANY

A company is a legal business entity formed by a group of individuals to engage in and operate a commercial or industrial enterprise. It is recognized by law as a separate entity from its owners, with its own rights, responsibilities, assets, and liabilities.

DEFINITION OF COMPANY

"A company is defined as an artificial person, invisible, intangible created by or under law, with a discrete legal capacity, perpetual succession and a common seal"

By Chief justice Marshall

CHARACTERISTICS OF COMPANY

- ✓ **Incorporate association**

A company is created when it is registered under the company's act. It comes into being from the date mentioned in the certificate of incorporation. It may be noted in this connection that section 11 provided that an association of more than ten persons carrying on business in banking or an association or more than twenty persons carrying on any other type of business must be registered under the companies Act.

- ✓ **Separate Legal Entity**

A company has a legal distinct entity and is independent of its members. The creditors of the company can recover their money only from the company and the property of the company. They cannot sue individual members. Similarly, the company is not in any way liable for the individual debts of its members. The property of the company is to be used for the benefits of the company and nor for the personal benefits of the shareholders.

- ✓ **Common Seal**

A company being an artificial person has no body similar to natural person and as such it cannot sign documents for itself. It acts through natural person who are called its directors. But having a legal personality. It can be bound by only those documents which bear its signature.

- ✓ **Limited liability**

A company may be limited by shares or a company limited by guarantee. In company limited shares, the liability of members is limited to the unpaid value of the shares.

- ✓ **Transferable Shares**

In a public company, the shares are freely transferable. The right to transfer shares is a statutory right and it cannot be taken away by a provision in the articles. The articles shall prescribe the manner in which such transfer of shares will be made and it may also contain bona fide and reasonable restrictions on the right of members to transfer their shares.

- ✓ **Separate property**

As a company is a legal person distinct from its members, it is capable of owning, enjoying and disposing of property in its own name. Although its capital and assets are contributed by its shareholders, they are not the private and joint owners of its property. The company is the real person in which all its property is vested and by which it is controlled, managed and disposed of.

- ✓ **Delegated Management**

A Joint Stock company is an autonomous, Self-governing and Self-Controlling Organization. Since it has a large number of members, all of them cannot take part in the management of the affairs of the company. Actual control and management is delegated by the shareholders to their elected representatives as directors.

- ✓ **Voluntary Association for Profit**

A company is formed Voluntary by individuals or entities with the intention of conducting business for profit. Profits are shared among shareholders as dividends.

OBJECTIVES OF THE STUDY

PRIMARY OBJECTIVES

To study on Impact of predictive Analysis for companies engaging performance after the pandemic in BBK Exports Private Limited.

SECONDARY OBJECTIVES

- To understand the role of predictive analysis in post pandemic business environments.
- To evaluate the effectiveness of predictive analytics on business performance.
- To identify the challenges companies, face in implementing predictive analytics post pandemic.
- To examine industry specific impact of predictive analytics, post COVID- 19.
- To assess the contribution of predictive analytics in strategic decision-making.
- To analyze the return on investment (ROI) from predictive analytics solutions.
- To explore how predictive analytics influences customer engagement and satisfaction.
- To provide recommendations for improving predictive analysis practices.

NEEDS OF STUDY

The COVID-19 pandemic disrupted global markets, altering supply chains, customer behavior, employee's management, and overall business operations. In the post pandemic era, predictive analysis has become a critical tool for companies aiming to regain stability, boost performance, and prepare for future uncertainties.

- Managing Uncertainty and Risk
- Enhancing Decision-making
- Forecasting Market Trends
- Improving Operational Efficiency
- Supporting Digital Transformation
- Better Customer Relationship Management
- Enhancing Financial Performance
- Competitive Advantages

SCOPE OF STUDY

- The scope outline how predictive analytics can be applied across various areas of business performance post pandemic.
- Predictive analytics is relevant across multiple industries like, Logistics and supply chain, Healthcare, Banking and Financial Services.
- Functional Scope within Organization such as, Marketing Analytics, Sales Forecasting, Supply chain Optimization, Financial Planning.
- Customer Behavior and Market Dynamics helps to understand and adopt the markets dynamics.
- Technological Integration with AI, machine learning, big data expands the capabilities of predictive analysis.

LIMITATION OF STUDY

While predictive analytics offers many benefits for improving company performance after the pandemic, there are several limitations and challenges that organization must consider.

- ✓ Data Availability and Accuracy
- ✓ High Implementation cost
- ✓ Changing Customer Behavior
- ✓ Model limitation and Assumptions
- ✓ Security and Privacy concerns
- ✓ Dependence on External Variables

INDUSTRY PROFILE



Industry type:

Leather and Footwear Manufacturing & Export Industry. BBK Exports private Limited operates within India's leather and footwear export industry, a vital part of the country's consumer goods and manufacturing sector. It primarily specialized in the production and export of leather footwear components, finished leather goods, and accessories. The company is based in Tamil Nadu, a major leather processing and manufacturing hub in India.

Industry Overview:

The global leather goods market is valued at over USD 400 billion, with major demand from Europe, North America, and East Asia. India is among the top 5 exporters of leather goods and footwear globally. India exports leather products to more than 100 countries, and Tamil Nadu accounts for nearly 40% of India's leather exports.

Footwear Expected to reached USD 550+ billion by 2027, growing at CARG of ~4.3%.

Leading Producers:

- **China:** World's largest producer and exporter of leather footwear.
- **Vietnam, Italy, Indonesia, India:** They are also a key player in leather goods manufacturing.
- **Italy:** Dominates premium/luxury leather export.



Industry Outlook (2025-2030)

Aspects	Outlook
Exports Growth	Targeting USD 10 billion by 2030 (CLE target)
FDI Inflow	Growing, especially in fashion/luxury footwear.
Technology Adoption	CAD/CAM,AI in design, automation
Sustainable Practices	Shift to chrome free tanning, water reuse

COMPANY PROFILE

BBK Exports Private Limited is a Tamil Nadu based manufacturer and exporter specializing in leather goods and footwear components. The company has earned recognition in both domestics and international markets for producing high-quality leather products, primarily serving the fashion, footwear, and accessory sectors.

The company maintains a strong products portfolio focused on quality craftsmanship, export compliance, and customization. Their specialization in leather shoe uppers and premium leather accessories has positioned them as reliable partner in the global leather trade.

BBK operates tanneries and footwear production units in Tamil Nadu, mainly in Ranipet and Kanchipuram districts, which are part of India's "Leather Cluster".

Name: BBK Exports Private Limited.

Timing: 8.30 A.M to 5.00 P.M

Working Day: Monday to Saturday

Address: No:21, Kilambi village, Krishnapuram Post, Kanchipuram-631531.

Company Status

- Registrar of the company: R.O.C Chennai.
- BBK exports private limited is a private limited company incorporated on 25th January 2006.
- It is classified as a Non-Government Company.
- In the year 2017, a company has started in the name of BBK exports private limited company.
- It is a leading manufacturing and exports of higher quality leather products.
- The company located at the place, that has man power.
- Directors of BBK exports private limited are Narayanasamy Baskaran, Porselvi Baskaran, Barathkumar Baskaran, Loganathan Keerthi Varman.
- BBK exports private Limited's annual general meetings(AGM) was held on the N/A and as per records of ministry of corporate affairs(MCA) its balance sheet was last field on 31sr March of every year.

Vision of the company

To produce the best quality finished leather maintaining international standards with the state of the art technology.

Mission of the company

- ✓ The first mission is to promote and save their products.
- ✓ They represent their product outside the country.
- ✓ They are working carefully because their product cannot harm environment.
- ✓ Ensuring a positive customer experience, making available goods and services that are top-notch quality.

FACTORY LICENSE:

- Factory owners had to compulsorily register their premises with local authorities before the Factories Act, 1948 come into force.
- The Factories Act, 1948 provides that certain factories that fulfill the conditions given in the act must register and obtain a factory license under the act to ensure the safety of the workers.
- **Legal assurance:** Enhances the safety of the workplace for employees by ensuring to safety standards.
- **Legal Compliance:** Ensures that the factory operates within the legal framework, avoiding penalties and legal issues.

MACHINARIES USED IN THE FACTORY



FUSING MACHINE

- A Fusing machine is a piece of equipment that uses heat and pressure to bond fabrics and interlining materials together.
- Uses low temperature and high pressure technology to fuse leather evenly without causing it to lose moisture.

PLATTING CUTTING MACHINE

- A leather plate embossing machine is a hydraulic press that can cut and emboss leather for a variety of applications.
- These machines can achieve efficient, precise and repeatable cutting process through different cutting methods and processes.

SEMI-ELECTRIC STACKER

- Semi-Electric stacker is a type of material handling equipment used for lifting, move and lower loads.
- Its function is to pile bilk material such as limestone, ores, coal and cereals onto a stockpile.

PRESSING MACHINE

- Pressing machines used in the leather industry includes belt embossing machines, hydraulic baling presses, and hydraulic plane leather belt embossing machines.
- Embossing on leather is pressing a design into the leather, creating a beautiful and unique look. It can create intricate patterns that can add style to any project.

BELT CONVEYOR MACHINE

- Belt conveyors are used in many industries, including manufacturing, logistics, and food processing.
- They are also used in the leather industry to move materials efficiently and safely.

BEAM CUTTING PRESS

- A beam cutting press is a hydraulic machine that can cut leather and other material with precision.
- It uses hydraulic pressure to cut materials with consistent force.

TRANSPORTATION MANAGEMENT

Transportation management in office management involves planning, coordinating, and implementation transportation activities for a company. It includes managing the fleet of vehicles, optimizing logistics networks, and developing transportation policies.



Transportation management activities

- **Fleet management:** Managing the company's fleet of vehicles, including purchasing, leasing, maintenance, and repairs.
- **Operational planning:** Optimizing the logistics network to fulfill transportation requirements.
- **Budget management:** Controlling the company's budget, overseeing performance, and planning for changes to schedules and policies.
- **Capacity planning:** Planning and executing logistics to maximize savings and use resources efficiently.

EMPLOYEES WORK AT BBK EXPORTS

BBK Exports has a workforce of 650 employees as of April 07, 2024.

Export: Based on the historical data available with Volza, the company has exported 5,807 shipments to 187 buyers. The primary markets for these exports were United Kingdom, the United Stated, and Canada. Key exported products fell under HSN codes 6406, 4202, and

6403. The major buyers involved were Beaconsfield Footwear Ltd, Coach Services Inc Dha Kate Spade, and Coach Services Care of Nis.

Imports: As per the historical data from Volza, BBK Exports Limited imported 2,923 shipments from 190 suppliers. The main sources for these imports were China, the Vietnam, and Italy. The imported goods primarily consisted of items categorized under HSN codes 4107, 9607, and 5904. The leading suppliers for these imports were Coronet Vietnam Co Ltd, Gruppo Mastrotto Spa, and Avery Dennison Hong Kong B V.

PRODUCT PROFILE

BRANDS

COACH









MAKING PRODUCTS

- Shoes
- Wallet
- Belt
- Handbags
- Bags



REVIEW OF LITERATURE
CHAPTER - 2

Keplercannon (2020):

Analytics terms recalibrated models due to pandemic shifted data patterns, decision making required shorter time windows and fresh feature engineering. A strategy and analytics consulting firm, published a report in 2020 addressing how predictive analytics played a crucial role in helping companies navigate the uncertainty caused by the COVID-19 pandemic.

HashStudioz (2025):

Business intelligence and data mining were essential for post-pandemic recovery and growth across industries. A leading technologies and AI solution firm, released a white paper in 2025 detailing how predictive analytics transformed business performance in the aftermath of the pandemic. The study focused on real- world applications and success stories across multiple industries.

FreightAmigo (April 2025):

Predictive analytics optimized inventory and demand forecasting reducing excess inventory by 25% and stock outs by 40%. A global logistics technology platform, released a comprehensive report emphasizing the critical role of predictive analytics in optimizing logistics and supply chain operations post pandemic. The study evaluated how data-driven intelligence influenced decision-making and performance in logistics dependent businesses.

Kinkedpress (2024):

Real time AI/ML adoption surged for timely decision making and resilience amid volatility. An emerging research publisher in business and data science, published a through leadership article in 2024 focused on the growing role of predictive analytics in enhancing enterprise performance after the pandemic. The publication included cross-industry surveys, case studies, and expert interviews.

McKinsey (Sept 2020):

Many analytics models failed due to assumption breaks in pandemic data, leaders had to rebuild forecasting frameworks entirely. Highlights that machine learning models built on pre-pandemic data lost predictive power during COVID-19 changing behaviors lockdowns, travel

bans, shifts to digital undermined underlying patterns, rendering many models unreliable or misleading.

Tiger analytics (Dec 2022):

Analytics leaders adopted cloud platforms and co-innovation to drive agility and resilience. Through the blog does not share quantitative ROI figures, it posits that companies embracing these practices gained agility, mitigated risk, and optimized decision-making through enhanced predictive capabilities.

Economic Times CIO (2024):

Predictive analytics embedded in business strategy enables proactive rather than reactive response to market volatility. With special emphasis on the Economic Times CIO (ETCIO) article from May 2021 and related newer studies and commentary.

Binary Semantics:

Predictive analytics in manufacturing improved efficiency up to 25%, quality by 20-30%, and supply chain efficiency by 15-30%. Offers predictive analytics and business intelligence platforms utilized across industries like finance, healthcare, retail, manufacturing, logistics, and marketing.

Londhe et al., JISEM:

Predictive analytics enhances employee's performance evaluation and succession planning by increasing fairness and accuracy. In regulated sector like pharma, firms had to re-evaluate which historical data still holds validity for distinguished between pandemic altered behaviors and enduring trends.

Wamba & team:

Big data and predictive capabilities improve flexibility, agility, and overall firm performance especially in supply chain. In a 2015 systematic review and longitudinal case study of demonstrated that organizations with high analytics maturity achieve 5-20% improvements in performance metrics such as operational efficiency and cost reduction.

Schmitt (2022):

AutoML frameworks help firms quickly deploy predictive models and close skills gaps. It appears that Marc Schmitt's 2022 work largely focuses on methodological topics such as Deep learning in business analytics rather than empirical studies on pandemic recovery performance.

Tawil et al. (2023):

SMEs using predictive analytics saw increased productivity, innovation, and customer service, despite financial adoption barriers. Malaysian hotels found big data analytics capabilities (BDAC) significantly improved organizational agility, marketing & organizational innovation, and firm performance in the post lockdown.

Kraus et al. (2018):

Deep learning boosts operational prediction accuracy beyond classical ML, enhancing performance in competitive environments. They demonstrated that deep learning models delivered superior predictive performance over traditional machine learning in multiple real world case studies, translating to enhanced operational performance for firms.

Stephany et al. (2020):

The CoRisk Index used real-time text mining to assess COVID-19 risks by industry, aiding firms and policymakers. Stephany and colleagues developed a data-mining and natural language processing (NLP) methods to analyze SEC filings and produce a real-time index (CoRisk-Index) that captures COVID-19 risk awareness and perceptions across industries.

TechRadarPro (2025):

97% of analysts now integrate AI, 87% use automation in analytics workflows and improving ROI visibility and decision impact. By automating metrics like time saved, cost reduction, and revenue attribution to analytics, generated insight, firms can demonstrate tangible performance benefits directly linked to predictive analytics.

TechRadarPro cybersecurity (2025):

71% executive report AI improved productivity post pandemic, though frontline skepticism persists, Despite AI-driven tools advancing rapidly, most organizations have adopted hybrid cloud architectures and are making trade-offs.

MoldStud survey (2024):

Businesses saw 20% lift in retention, 15% cost reduction, 30% lower turnover, and 25% fewer defaults with predictive tools. He emphasized that poor data quality directly reduces the effectiveness of predictive models impacting revenue, operational efficiency, and cost saving. Data governance and validation frameworks are thus foundational to performance.

Scholink Survey (2025):

Reported decision speed increase 73.3%, forecast accuracy increase 76.6%, and customer satisfaction increase 67.6% among predictive analytics adopters. Firms with greater adoption profiled savings of 60-70% in operational costs indicating that analytics not only enhance insight but also deliver substantial bottom-line performance benefits.

BARC Study (June 2025):

While 69% view predictive planning as vital, only 11% use it actively only 4% use AI in performance management. Firms using these integrated predictive planforms demonstrate enhanced ability to detect external shifts and respond swiftly crucial for post pandemic responsiveness.

Rai & Kumar (2021):

Firms using predictive models in marketing saw better targeting and customer retention post pandemic. That analytics are most beneficial when paired with dynamic organizational capabilities such as agility and innovation which mediate the impact on firm performance.

Kumar et al. (2022):

Predictive maintenance models improved equipment uptime and reduced costs in manufacturing settings. A china-based study (2025) analyzing manufacturing firms found that big data analytics (BDA) improved decision-making and resilience, particularly when integrated with circular economy practices and digital marketing strategies.

Deloitte report (2021):

HR analytics predicted employee attrition and skills gap in remote & hybrid workforce scenarios. Deloitte surveys reveal that CFOs and finance leaders prioritized strengthening forecasting and analytics skills post pandemic.

McKinsey & Co (2021):

Firms using predictive analytics enjoyed 20-25% efficiency gains in operations and logistics. Companies with a high "analytic quotient" significantly outperformed peers in growth and shareholders returns, highlighting the sustained value of predictive analytics maturity.

Ivanov & Dolgui (2020):

Supply chain simulations via predictive models supported scenario planning and faster recovery. Empirical studies in high-demand sectors show that supply chain agility, supported by information technology, decisively improves firm performance during COVID-19. Predictive modelling helps to dynamically adopt business strategy and maintain service levels especially in crisis sensitive industries.

Batra & Dey (2023):

Companies combining predictive insights with strategy showed faster bounce-back, higher customer satisfaction, and competitive advantage. A notable study in technological forecasting and social changes investigated the joint impact of big data analytics on decision-making, forecasting, and firm performance across Indian firms.

RESEARCH METHODOLOGY
CHAPTER - 3

Technology and customer taste and preference play a vital role in today's generation. Research methodology is a set of various methods to be followed to find out various information's regarding market strata of different products. Research methodology is required for every industrial service industry for getting acquire knowledge of their products.

RESEARCH DESIGN

Descriptive and Analytical Research are made. To study how predictive analytics has influenced the operational and financial performance of company's post-pandemic, particularly in decision-making, risk mitigation, and strategic planning. It is a preliminary study based on primary data and the findings can be consolidated after a detailed conclusive study has been carried out.

RESEARCH INSTRUMENTS

The Research Instruments has Questionnaires.

SAMPLE DESIGN

Companies that adopted or expanded predictive analytics in post pandemic. Specifically, across sectors like manufacturing, retail, logistics, finance, and IT. Middle and senior-level employees involved in analytics, decision-making, or IT (e.g., data analysts, CIOs, business managers).

SAMPLE PROCEDURE

In the study "Convenience Sampling procedure is used. Convenience sampling is preferred because of some limitation and the complexity of the random sampling. Area

sampling is used in combination with convenience sampling so as to collect the data from different regions of the city and to increase reliability.

SAMPLE SIZE

Number of samples is = 100

PERIOD OF STUDY

The study was conducted over a period of 30 days during the MBA internship.
Duration: 01 July 2025 to 30 July 2025

DATA COLLECTION

Data is collected from various employees through personal interaction. Specific questionnaire is prepared for collecting data. Data is collected with mere interaction and formal discussion with different respondents.

SOURCE OF DATA

- Primary data
- Secondary data

PRIMARY DATA

The data collected for the first time through observation and interview method. The data is collected by observing the working of various departments and also by interview the manager of the entire department. It is also obtained by the help of staff members.

SECONDARY DATA

The data is collected by secondary sources also. The data is collected through company manual, product brochure, company website and annual report.

TOOLS AND TECHNIQUES

The investigator selected questionnaires survey as a tool for the purpose of data collection the survey schedule was the prepared relations of the following areas.

This depends on the purpose for which the survey is conducted. It the purpose is to study the entire universe of a variable them only census method of data collection should be adopted.

- Charts
- Percentage analysis

CHART:

Bar charts, line chart, column chart, pyramid chart, doughnut chart, cylinder chart and pie charts are used for analysis to get a clear idea about the tabulated data.

PERCENTAGE ANALYSIS:

Increase the percentage analysis percentage is calculated by much playing the respondents into100 and it is divided sampling size.

$$\text{Percentage} = \frac{\text{No. of Respondents}}{\text{Sampling size}} \text{X100}$$

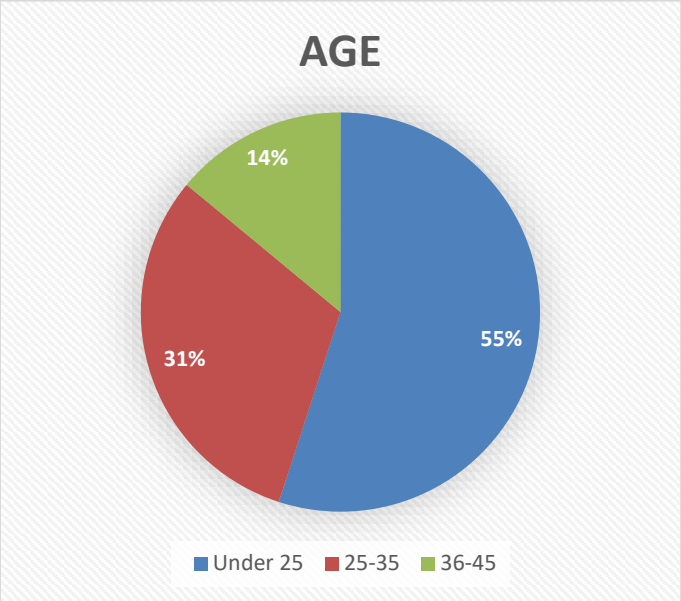
DATA ANALYSIS AND INTERPRETATION
CHAPTER - 4

TABLE NO: 01

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF AGE

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Under 25	55	55%
02	25-35	31	31%
03	36-45	14	14%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF AGE



INFERENCE

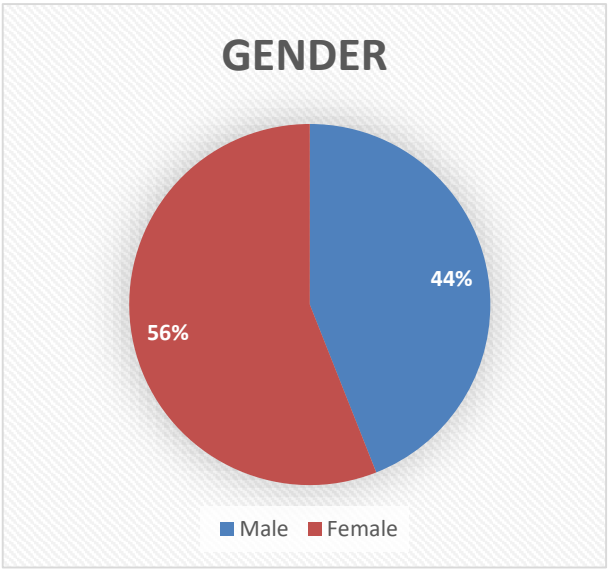
From the above table indicates that, out of 100 respondents, 55% of Under 25 age, 31% of 25-35 age and 14% of 36-45 age. It is to conclude that majority of 55 respondents of the total were under 25 age study.

TABLE NO: 02

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF GENDER

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	**Male**	**44**	**44%**
02	**Female**	**56**	**56%**
03	**Others**	**Nil**	**Nil**
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF GENDER



INFERENCE

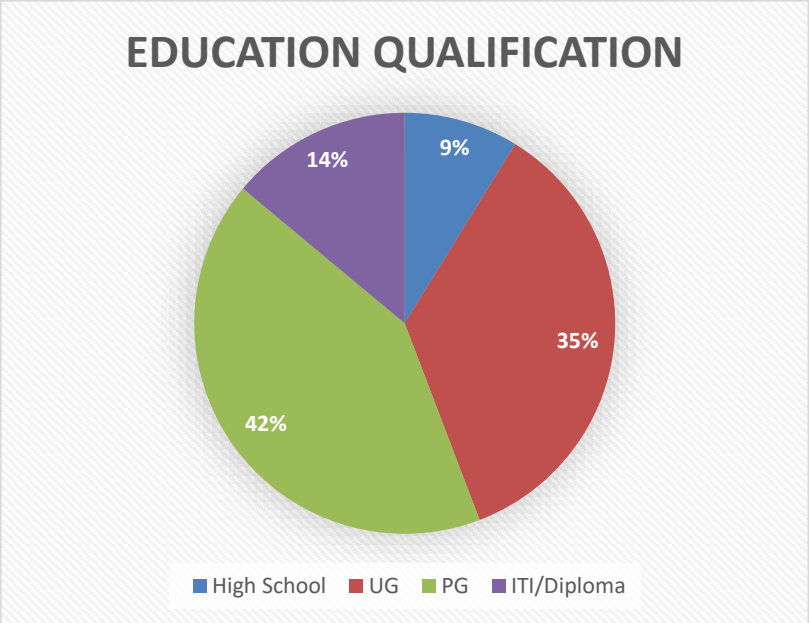
From the above table indicates that, out of 100 respondents, 44% of Male and 56% of Female. It is to conclude that majority of 56 respondents of the total were under Female study.

TABLE NO: 03

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF EDUCATION QUALIFICATION

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	High School	15	15%
02	UG	33	33%
03	PG	39	39%
04	ITI/Diploma	13	13%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF EDUCATION QUALIFICATION



INFERENCE

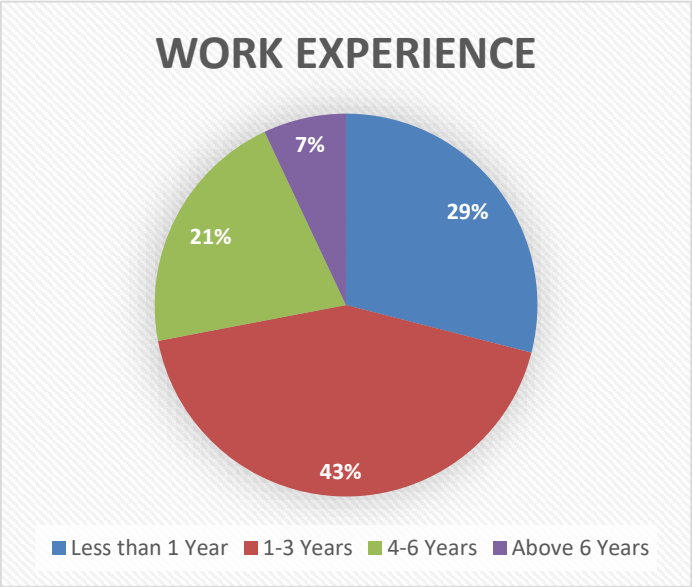
From the above table indicates that, out of 100 respondents, 9% of High School, 35% of UG, 42% of PG and 14% of ITI/Diploma. It is to conclude that majority of 42 respondents of the total were under PG study.

TABLE NO: 04

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF WORK EXPERIENCE

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Less than 1 year	29	29%
02	1 – 3 years	43	43%
03	4 – 6 years	21	21%
04	Above 6 years	7	7%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF WORK EXPERIENCE



INFERENCE

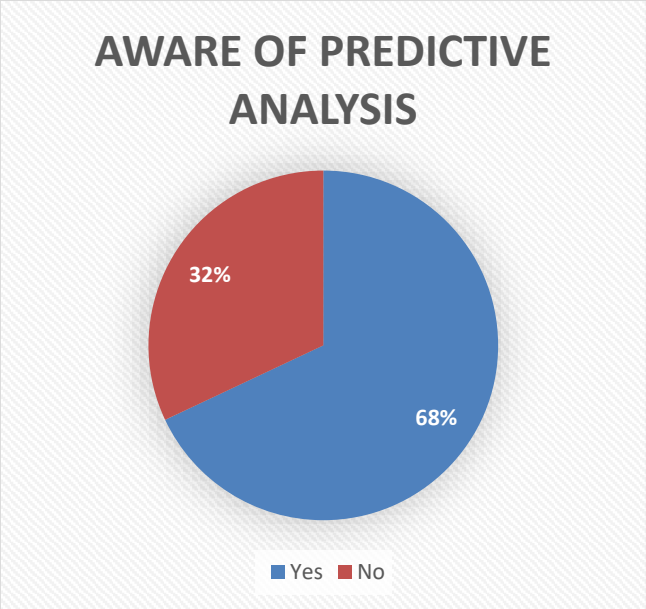
From the above table indicates that, out of 100 respondents, 29% of Less than 1 year, 43% of 1-3 Years, 21% of 4-6 Years and 7% of Above 6 Years. It is to conclude that majority of 43 respondents of the total were under 1-3 Years of Work Experience study.

TABLE NO: 05

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF AWARE OF PREDICTIVE ANALYSIS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Yes	68	68%
02	No	32	32%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF AWARE OF PREDICTIVE ANALYSIS



INFERENCE

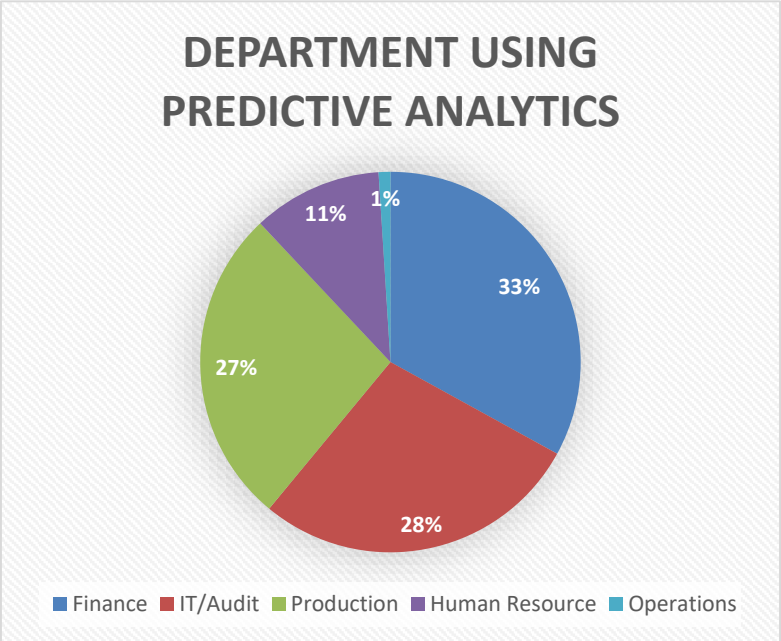
From the above table indicates that, out of 100 respondents, 68% of Yes and 32% of Agreed. It is to conclude that majority of 68 respondents of the total were under Aware of Predictive analysis study.

TABLE NO: 06

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF DEPARTMENTS USING PREDICTIVE ANALYTICS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Finance	33	33%
02	IT/Audit	28	28%
03	Production	27	27%
04	Human Resource	11	11%
05	Operations	1	1%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF DEPARTMENTS USING PREDICTIVE ANALYTICS



INFERENCE

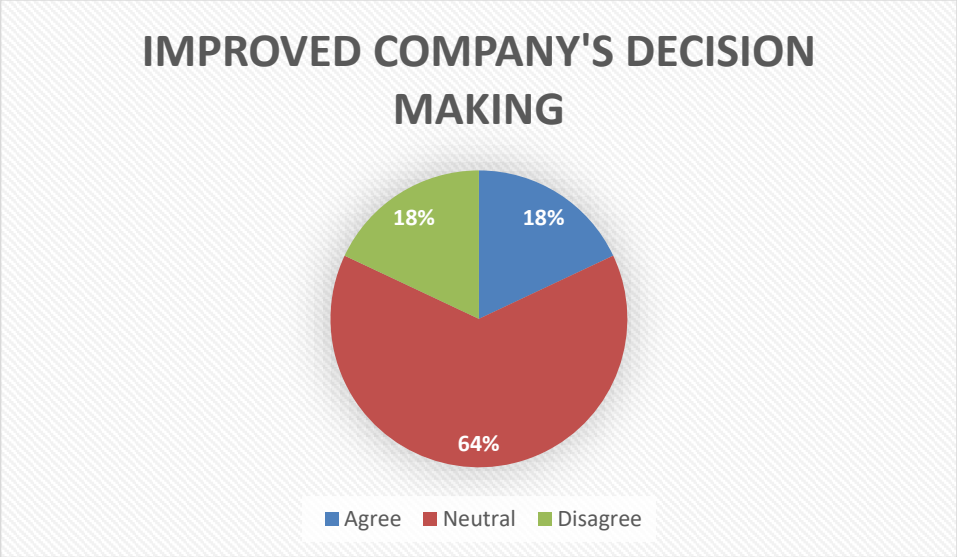
From the above table indicates that, out of 100 respondents, 33% of Finance, 28% of IT/Audit, 27% of Production, 11% of Human Resource and 1% of Operations. It is to conclude that majority of 33 respondents of the total were under Finance study.

TABLE NO: 07

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF IMPROVED COMPANY'S DECISION MAKING

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Agree	18	18%
02	Neutral	64	64%
03	Disagree	18	18%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF IMPROVED COMPANY'S DECISION MAKING



INFERENCE

From the above table indicates that, out of 100 respondents, 18% of Agreed, 64% of Neutral and 18% of Disagreed. It is to conclude that majority of 64 respondents of the total were under Neutral study.

TABLE NO: 08

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF LEARN ABOUT PREDICTIVE ANALYSIS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Training	32	32%
02	Manager	37	37%
03	Internet	22	22%
04	Don’t Know	9	9%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF LEARN ABOUT PREDICTIVE ANALYSIS



INFERENCE

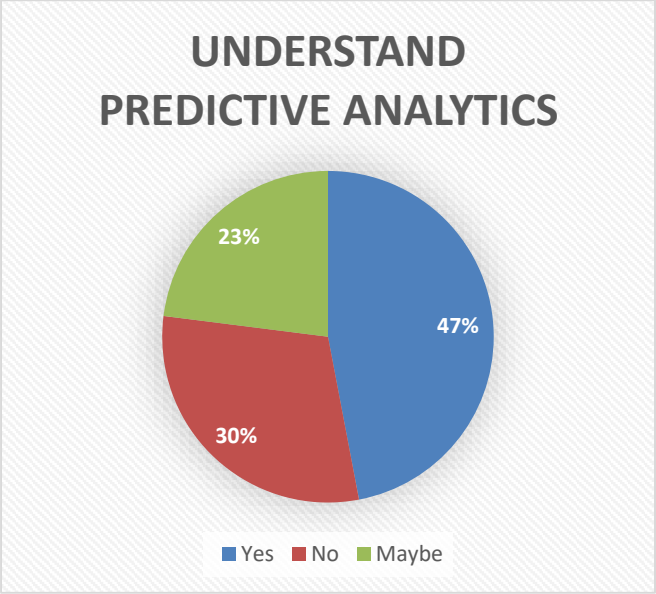
From the above table indicates that, out of 100 respondents, 32% of Training, 37% of Manager, 22% of Internet and 9% of Don’t Know. It is to conclude that majority of 37 respondents of the total were under Manager study.

TABLE NO: 09

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF UNDERSTAND PREDICTIVE ANALYTICS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Yes	47	47%
02	No	30	30%
03	Maybe	23	23%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF UNDERSTAND PREDICTIVE ANALYTICS



INFERENCE

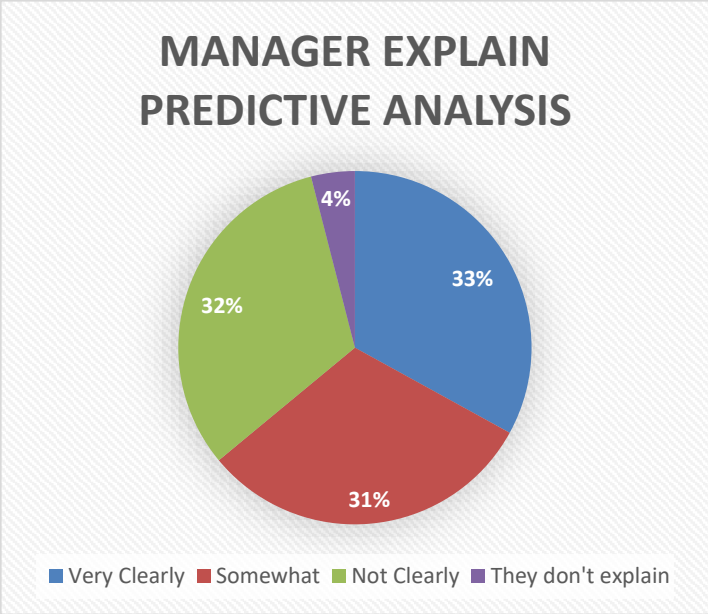
From the above table indicates that, out of 100 respondents, 47% of Yes, 30% of No and 23% of Maybe. It is to conclude that majority of 47 respondents of the total were under Predictive Analytics study.

TABLE NO: 10

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF MANAGER EXPLAIN PREDICTIVE ANALYSIS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Very Clearly	33	33%
02	Somewhat	31	31%
03	Not Clearly	32	32%
04	They don't explain	4	4%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF MANAGER EXPLAIN PREDICTIVE ANALYSIS



INFERENCE

From the above table indicates that, out of 100 respondents, 33% of Very Clearly, 31% of Somewhat, 32% of Not Clearly and 4% of They don't explain. It is to conclude that majority of 33 respondents of the total were under Manager Explain Very Clearly study.

TABLE NO: 11

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF HELPS TO AVOID BUSINESS RISKS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Yes	57	57%
02	No	26	26%
03	Maybe	17	17%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF HELPS TO AVOID BUSINESS RISKS



INFERENCE

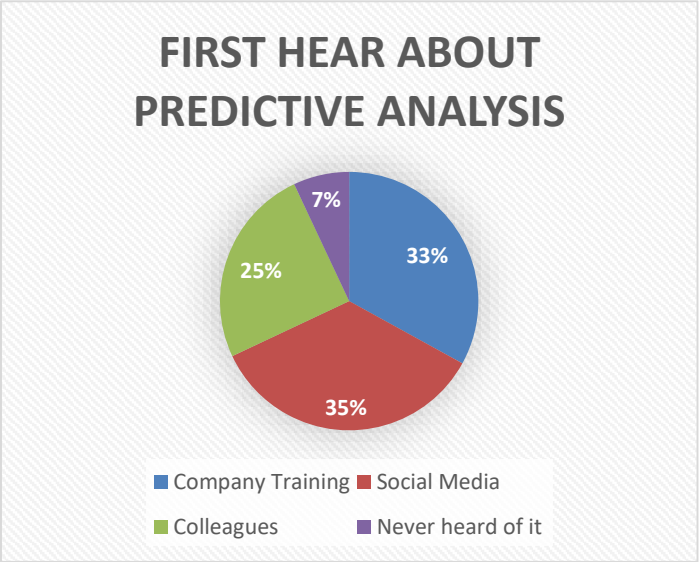
From the above table indicates that, out of 100 respondents, 57% of Yes, 26% of No and 17% of Maybe. It is to conclude that majority of 57 respondents of the total were under Helps to Avoid Business Risks study.

TABLE NO: 12

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF FIRST HEAR ABOUT PREDICTIVE ANALYSIS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Company Training	33	33%
02	Social Media	35	35%
03	Colleagues	25	25%
04	Never Heard of it	7	7%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF FIRST HEAR ABOUT PREDICTIVE ANALYSIS



INFERENCE

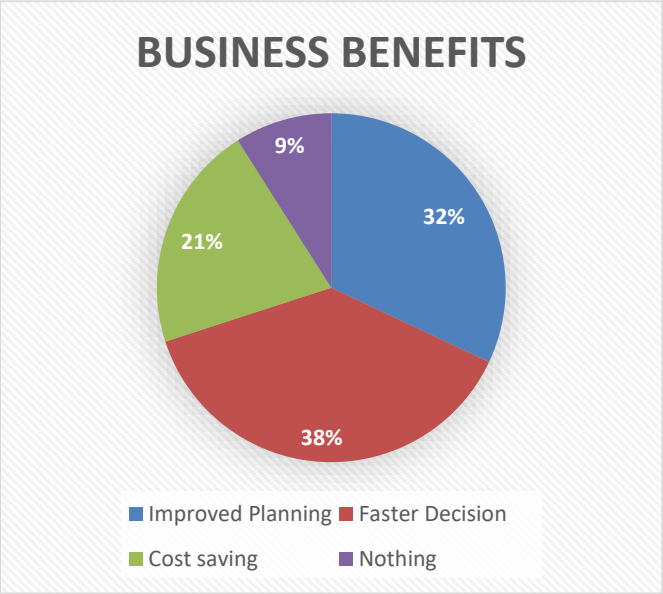
From the above table indicates that, out of 100 respondents, 33% of Company Training, 35% of Social Media, 25% of Colleagues and 7% of Never heard of it. It is to conclude that majority of 35 respondents of the total were under Social Media study.

TABLE NO: 13

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF BUSINESS BENEFITS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Improved planning	32	32%
02	Faster Decision	38	38%
03	Cost Saving	21	21%
04	Nothing	9	9%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF BUSINESS BENEFITS



INFERENCE

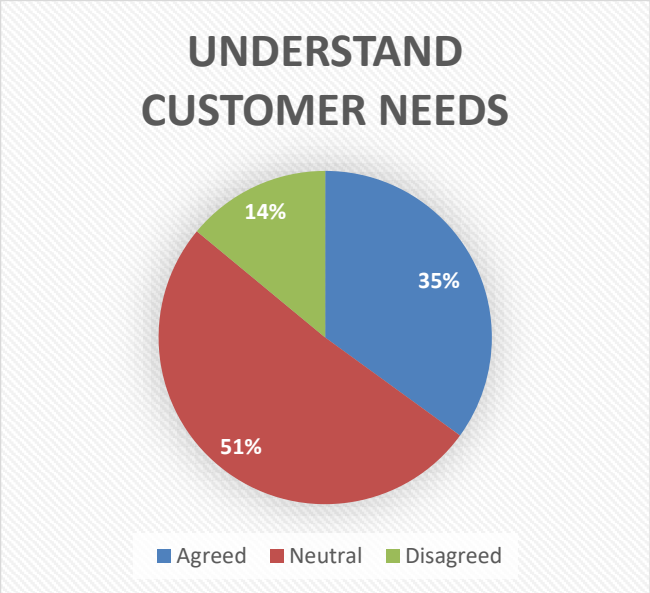
From the above table indicates that, out of 100 respondents, 32% of Improved Planning, 38% of Faster Decision, 21% of Cost Saving and 9% of Nothing. It is to conclude that majority of 38 respondents of the total were under Faster Decision study.

TABLE NO: 14

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF UNDERSTAND CUSTOMER NEEDS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Agreed	35	35%
02	Neutral	51	51%
03	Disagree	14	14%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF UNDERSTAND CUSTOMER NEEDS



INFERENCE

From the above table indicates that, out of 100 respondents, 35% of Agreed, 51% of Neutral and 14% of Disagreed. It is to conclude that majority of 51 respondents of the total were under Neutral study.

TABLE NO: 15

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF IMPROVED CUSTOMER SATISFACTION

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Yes	46	46%
02	No	24	24%
03	Not Sure	30	30%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF IMPROVED CUSTOMER SATISFACTION



INFERENCE

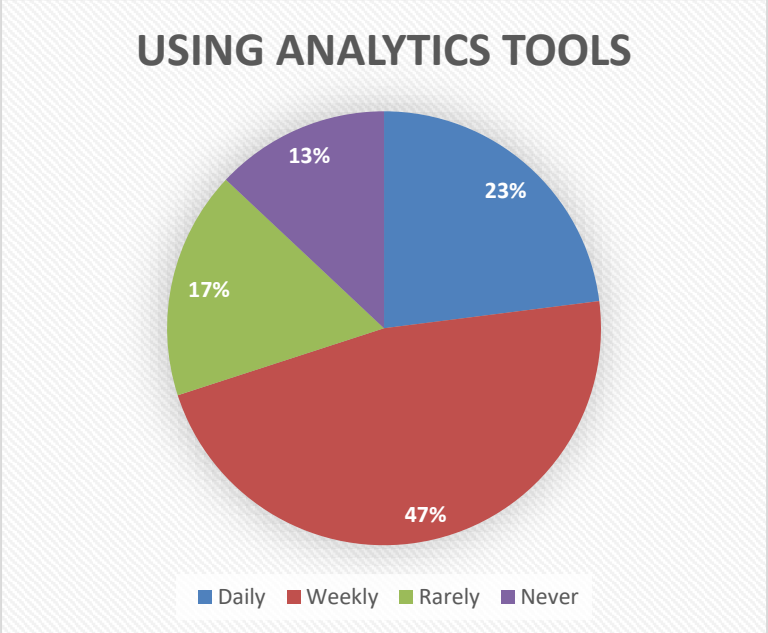
From the above table indicates that, out of 100 respondents, 46% of Yes, 24% of No and 30% of Not Sure. It is to conclude that majority of 46 respondents of the total were under Improved Customer Satisfaction study.

TABLE NO: 16

CLASSIFICATION OF RESPONDENTS ON THE BASIS OF USING ANALYTICS TOOLS

S.NO	PARTICULARS	NO.OF.RESPONDENTS	PERCENTAGE
01	Daily	23	23%
02	Weekly	47	47%
03	Rarely	17	17%
04	Never	13	13%
	TOTAL	**100**	**100%**

CHART REGARDING RESPONDENTS ON THE BASIS OF USING ANALYTICS TOOLS



INFERENCE

From the above table indicates that, out of 100 respondents, 23% of Daily, 47% of Weekly, 17% of Rarely and 13% of Never. It is to conclude that majority of 47 respondents of the total were under Weekly study.

SUMMARY OF FINDINGS, SUGGESTION AND CONCLUSION
CHAPTER - 5

FINDINGS:

- ✓ Many companies previously relied on historical averages and traditional forecasting methods that proved inadequate during the pandemic due to sudden market shocks and volatile customer behaviors.

- ✓ Predictive models have optimized resource allocation and operational planning. By predicting peak periods of customer demand, labor shortages, or logistical delays, companies have reduced operational costs and improved efficiency.

- ✓ After the pandemic, companies faced rising risks such as financial fraud, cybersecurity threats, and supply chain vulnerabilities. Predictive analytics played a significant role in identifying anomalies and patterns of fraudulent activity.

- ✓ Predictive analytics allowed businesses to segment customers more precisely, anticipate churn, and personalize marketing campaigns.

- ✓ Companies that leveraged predictive analytics reported faster and more confident decision-making. With reliable forecasts, leaders were able to develop resilient strategies, such as adjusting pricing, exploring digital channels, and diversifying supply chains.

- ✓ The impact of predictive analytics was not limited to a single sector. Manufacturing companies used predictive maintenance to reduce machinery downtime, IT firms adopted analytics to predict cybersecurity threats, banks deployed models to forecast loan defaults, and healthcare organizations forecasted disease outbreaks.

- ✓ Despite benefits, companies faced challenges: lack of skilled professionals, high costs of data infrastructure, data privacy concerns, and resistance from employees unfamiliar with advanced technology.

SUGGESTIONS:

- ✓ Companies must upgrade their IT systems to handle large volumes of structured and unstructured data. Cloud-based platforms, artificial intelligence (AI), and machine learning (ML) tools can provide scalability, real-time analytics, and cost-efficiency.

- ✓ Training and upskilling employees in data literacy, analytics, and interpretation are crucial. Companies should collaborate with academic institutions or offer in-house training programs to bridge the skill gap.

- ✓ Reliable predictive analysis depends on clean, consistent, and high-quality data. Companies should implement strong data governance policies that address accuracy, security, and compliance.

- ✓ Organizational culture must shift from intuition-based decision-making to evidence-based strategies. Leaders should promote the use of analytics in all decision-making processes and encourage cross-departmental collaboration.

- ✓ Manufacturing firms should focus on predictive maintenance, financial institutions on fraud detection, retailers on demand forecasting, and healthcare on patient care predictions. Customization ensures relevance and maximizes impact.

- ✓ Post-pandemic markets remain uncertain; thus, real-time data processing is essential. Companies should implement real-time dashboards and AI-driven alerts to respond swiftly to market changes, supply chain disruptions, or customer behavior shifts.

- ✓ Smaller businesses with limited budgets should adopt cost-friendly predictive tools, such as open-source platforms or subscription-based analytics services.

- ✓ Predictive analytics should not function in isolation but be integrated with corporate strategy, risk management, and performance measurement systems.

CONCLUSION:

The study highlights that predictive analysis has emerged as a powerful enabler for companies striving to rebuild and thrive after the pandemic. The disruptions caused by COVID-19 exposed the limitations of traditional forecasting methods, pushing organizations to adopt more advanced analytical approaches. Predictive analytics not only helped companies manage uncertainty but also provided them with competitive advantage in an unpredictable environment.

By accurately forecasting demand, optimizing resources, and detecting risks, predictive analysis has improved operational efficiency and financial performance across industries. More importantly, it has allowed companies to remain customer-centric by anticipating evolving needs and delivering personalized experiences. Firms that successfully integrated predictive analytics into their strategies have shown higher resilience, adaptability, and innovation.

However, challenges such as skill shortages, high costs, data privacy issues, and cultural resistance cannot be ignored. Addressing these barriers is essential for the broader adoption of predictive analytics. The future success of organizations will depend not just on possessing data but on their ability to transform that data into actionable insights through predictive models.

The post-pandemic world is characterized by volatility, uncertainty, complexity, and ambiguity (VUCA). In such an environment, predictive analysis serves as a guiding compass, allowing businesses to prepare for disruptions, mitigate risks, and identify opportunities ahead of competitors.

In conclusion, predictive analytics is no longer optional but a strategic necessity. Companies that embrace data-driven decision-making will not only recover from the pandemic's impact but also chart a path toward sustainable growth, innovation, and long-term success. Those that ignore it risk falling behind in an era where information is the most valuable asset.

QUESTIONNAIRE

A STUDY ON IMPACT OF PREDICTIVE ANALYSIS FOR COMPANIES ENGAGING PERFORMANCE AFTER THE PANDEMIC

Respected Sir / Madam,

Your Participation is highly appreciated. The information provided by you will be kept strictly confidential and will be used only for academic research purposes. Kindly read each question carefully and answer to the best of your knowledge. Your cooperation will contribute significantly to the success of this research.

1. Name* _____

2. Age*

    - Under 25
    - 25-35
    - 36-45

3. Gender*

    - Male
    - Female
    - Others

4. Education Qualification*

    - High School
    - UG
    - PG
    - ITI/Diploma

5. Work Experience*

    - Less than 1 year
    - 1-3 Years
    - 4-6 Years
    - Above 6 Years

6. Are You Aware of Predictive Analysis? *

    - Yes
    - No

7. Did you Company start using predictive analysis after the pandemic? *

    - Yes
    - No
    - Not Sure

8. Which Department use predictive analytics the most? *

    - Finance
    - IT/Audit
    - Production
    - Human Resource

9. How often does your company use predictive analytics tools? *

    - Daily
    - Weekly
    - Rarely
    - Never

10. Has predictive analysis improved your company's decision making? *

- Agree
- Neutral
- Disagree

11. Has it helped identify new opportunities or trends post-pandemic? *

- Yes
- No
- Not sure

12. Do you feel training is required to understand predictive analytics? *

- Yes
- No
- Maybe

13. How did you learn about predictive analysis? *

- Training
- Manager
- Internet
- Don't Know

14. Predictive analysis helped avoid business risks? *

- Yes
- No
- Maybe

15. How clearly does your manager explain predictive analysis? *

- Very Clearly
- Somewhat
- Not Clearly
- They don’t explain

16. What business benefits has predictive analysis brought? *

- Improved Planning
- Faster Decision
- Cost Saving
- Nothing

17. Where did you first hear about predictive analysis? *

- Company Training
- Social Media
- Colleagues
- Never heard of it

18. Predictive analytics helped us understand customer needs better. *

- Agreed
- Neutral
- Disagreed

19. Did it improved customer satisfaction? *

- Yes
- No
- Not Sure

20. Should predictive analysis be a compulsory part of Business Strategy? *

- Yes
- No
- Not Sure

BIBLIOGRAPHY

- Davenport, T. H., & Harris, J. G. (2017). Competing on Analytics: The New Science of Winning. Harvard Business Review Press.

- Shmueli, G., Bruce, P. C., Gedeck, P., & Patel, N. R. (2020). Data Mining for Business Analytics: Concepts, Techniques, and Applications in R. Wiley.

- Manyika, J., et al. (2021). "The state of AI in 2021." McKinsey Global Survey. Retrieved from https://www.mckinsey.com

- Marr, B. (2016). Big Data in Practice: How 45 Successful Companies Used Big Data Analytics to Deliver Extraordinary Results. Wiley.

- Accenture. (2021). "How Predictive Analytics Can Help Businesses Thrive in the Post-Pandemic Era." Retrieved from https://www.accenture.com

- Chen, H., Chiang, R. H. L., & Storey, V. C. (2012). "Business Intelligence and Analytics: From Big Data to Big Impact." MIS Quarterly, 36(4), 1165–1188.

- Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). "From Data Mining to Knowledge Discovery in Databases." AI Magazine, 17(3), 37–54.

- PwC. (2022). "Data-driven recovery: Leveraging predictive analytics for post-pandemic business growth." Retrieved from https://www.pwc.com

- Brynjolfsson, E., & McElheran, K. (2016). "Data in Action: Data-Driven Decision Making and Predictive Analytics." Communications of the ACM, 59(5), 102–107.

- OECD. (2021). Impacts of COVID-19 on Global Trade and Supply Chains. OECD Publishing. Retrieved from https://www.oecd.org